KMeans++

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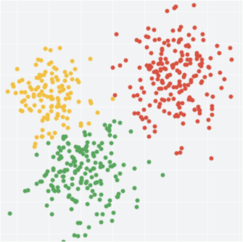

k-means++

For an object $\bar{X}$ in the dataset $\mathcal{D}$ let $R(\bar{X})$ be the distance from $\bar{X}$ to the closest cluster representative we have already chosen.

1. Select one representative $\bar{Y}_1$ uniformly at random from $\mathcal{D}$.
2. For every $i = 2, \ldots, k$
    2.1 Select representative $\bar{Y}_i$ from $\mathcal{D}$ with probability

$$\Pr(\bar{Y}_i = \bar{X}) = \frac{R(\bar{X})^2}{\sum_{\bar{X} \in \mathcal{D}} R(\bar{X})^2}$$

3. Proceed with the standard k-means using $\bar{Y}_1, \ldots, \bar{Y}_k$ as initial cluster representatives

k-means++

Within cluster sum of squares:

$$\phi = \sum_{x \in \mathcal{X}} \min_{c \in \mathcal{C}} \|x - c\|^2.$$

Theorem 3.1. *If $\mathcal{C}$ is constructed with k-means++, then the corresponding potential function ϕ satisfies*

$$\mathbb{E}[\phi] \leq 8(\ln k + 2)\phi_{\mathrm{OPT}}.$$

k-means++ vs k-means: experimental study

- Synthetic datasets Norm-10 and Norm-25.
  - Chose 10 (respectively 25) "real" centers uniformly at random from the hypercube of side length 500.
  - Then added points from a Gaussian distribution of variance 1, centered at each of the real centers.
  - Thus, we have a number of well separated Gaussians with the real centers providing a good approximation to the optimal clustering.

k-means++ vs k-means: experimental study

- Synthetic dataset Norm-10.

	Average ϕ		Minimum ϕ	
k	k-means	k-means++	k-means	k-means++
10	10898	5.122	2526.9	5.122
25	787.992	4.46809	4.40205	4.41158
50	3.47662	3.35897	3.40053	3.26072

Table: Experimental results on the Norm-10 dataset ($n = 10000$, $d = 5$)

Within cluster sum of squares: $\phi = \sum_{x \in \mathcal{X}} \min_{c \in \mathcal{C}} \|x - c\|^2$

k-means++ vs k-means: experimental study

- Synthetic dataset Norm-25.

	Average ϕ		Minimum ϕ	
k	k-means	k-means++	k-means	k-means++
10	135512	126433	119201	111611
25	48050.5	15.8313	25734.6	15.8313
50	5466.02	14.76	14.79	14.73

Table: Experimental results on the Norm-25 dataset ($n = 10000$, $d = 15$)

Within cluster sum of squares: $\phi = \sum_{x \in \mathcal{X}} \min_{c \in \mathcal{C}} \|x - c\|^2$

k-means++ vs k-means: experimental study

- The synthetic datasets: the k-means method does not perform well, because
  - the random seeding will inevitably merge clusters together, and the algorithm will never be able to split them apart.
  - the careful seeding method of k-means++ avoids this problem altogether, and it almost always attains the optimal results on the synthetic datasets

k-means++ vs k-means: experimental study

- Real datasets: Cloud dataset

k	Average ϕ		Minimum ϕ	
	k-means	k-means++	k-means	k-means++
10	7553.5	6151.2	6139.45	5631.99
25	3626.1	2064.9	2568.2	1988.76
50	2004.2	1133.7	1344	1088

Table: Experimental results on the *Cloud* dataset ($n = 1024, d = 10$)

Within cluster sum of squares:

$$\phi = \sum_{x \in X} \min_{c \in C} \|x - c\|^2.$$

k-means++ vs k-means: experimental study

- Real datasets: Intrusion dataset

k	Average ϕ		Minimum ϕ	
	k-means	k-means++	k-means	k-means++
10	$3.45 \cdot 10^8$	$2.31 \cdot 10^7$	$3.25 \cdot 10^8$	$1.79 \cdot 10^7$
25	$3.15 \cdot 10^8$	$2.53 \cdot 10^6$	$3.10 \cdot 10^8$	$2.06 \cdot 10^6$
50	$3.08 \cdot 10^8$	$4.67 \cdot 10^5$	$3.08 \cdot 10^8$	$3.98 \cdot 10^5$

Table: Experimental results on the *Intrusion* dataset $(n = 494019, d = 35)$

Within cluster sum of squares:

$$\phi = \sum_{x \in X} \min_{c \in C} \|x - c\|^2.$$

k-means++ vs k-means: experimental study

- The real-world datasets
- On the **Cloud** dataset, k-means++ terminates almost **twice as fast** while achieving potential function (within cluster sum of squares) values about **20% better**.
- On the larger **Intrusion** dataset: the potential value obtained by k-means++ is better by factors of **10 to 1000** and is also obtained **up to 70% faster**.